Lecture 4: Grouping and aggregation

TECH2: Introduction to Programming, Data, and Information Technology

Richard Foltyn
NHH Norwegian School of Economics

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See GitHub repository for notebooks and data:

https://github.com/richardfoltyn/TECH2-H26

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1 Grouping and aggregation with pandas

1.1 Aggregation and reduction

Similar to NumPy, pandas supports data aggregation and reduction functions such as computing sums or averages. By *"aggregation"* or *"reduction"* we mean summarizing multiple observations: for example, the mean reduces a series of observations (1 dimension) to a scalar value (0 dimensions).

pandas also provides built-in support for grouping data according to some criterion and applying these operations to each group.

The official user guide refers to these operations as *split-apply-combine* because they involve three steps:

1. *Split* data into groups based on some criteria;
2. *Apply* some function to each group separately; and
3. *Combine* the results into a single `DataFrame` or `Series`.

See also the pandas cheat sheet for an illustration of such operations.

We first set the path pointing to the folder which contains the data files used in this lecture.

[1]:
```
# Use files in the local data/ directory
DATA_PATH = '../../data'

# Uncomment this to load data directly from GitHub
# DATA_PATH = 'https://raw.githubusercontent.com/richardfoltyn/TECH2-H26/main/data'
```

1.1.1 Aggregations of whole Series or DataFrames

The simplest way to perform data reduction is to invoke the desired function on the entire `DataFrame`. We illustrate this using the Titanic dataset which we have encountered in the previous lectures.

We first load the data and tabulate the columns present in the `DataFrame`:

[2]:
```
import pandas as pd

# Path to Titanic passenger data CSV file
file = f'{DATA_PATH}/titanic.csv'

# Read in Titanic passenger data, set PassengerId column as index
df = pd.read_csv(file, index_col='PassengerId')

# Tabulate columns and number of observations
df.info(show_counts=True)
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 1 to 891
Data columns (total 9 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Survived  891 non-null    int64
 1   Pclass    891 non-null    int64
 2   Name      891 non-null    str
 3   Sex       891 non-null    str
 4   Age       714 non-null    float64
 5   Ticket    891 non-null    str
 6   Fare      891 non-null    float64
 7   Cabin     204 non-null    str
 8   Embarked  889 non-null    str
dtypes: float64(2), int64(2), str(5)
memory usage: 62.8 KB
```

We can now apply the `mean()` method to all numerical columns to compute the average for each column:

[3]:
```
# Compute mean of all numerical columns
df.mean(numeric_only=True)
```

[3]:
```
Survived     0.383838
Pclass       2.308642
Age         29.699118
Fare        32.204208
dtype: float64
```

Methods such as `mean()` are applied column-wise by default. The `numeric_only=True` argument selects numeric columns; without it, `mean()` raises an error for string columns.

Missing values (represented here by `np.nan`) are automatically ignored, as the following code demonstrates:

[4]:
```
import numpy as np

# mean() automatically drops missing observations
mean_pandas = df['Age'].mean()

# np.mean() returns NaN since some ages are missing (coded as NaN)
mean_numpy = np.mean(df['Age'].to_numpy())

print(f'Mean using Pandas: {mean_pandas:.1f}')
print(f'Mean using NumPy:  {mean_numpy:.1f}')
```

```
Mean using Pandas: 29.7
Mean using NumPy:  nan
```

NumPy also provides functions such as `np.nanmean()` that ignore NaNs.

> **Your turn**
>
> Use the Titanic dataset to perform the following aggregations:
>
> 1. Compute the median age using the column `Age` and the `median()` method.
> 2. Compute the fraction of female passengers using the information in column `Sex`.

1.1.2 Aggregations of subsets of data (grouping)

Applying aggregation functions to the entire `DataFrame` is similar to what we can do with NumPy. The added flexibility of pandas becomes obvious once we want to apply these functions to subsets of data, i.e., groups which we define based on column values or index labels.

For example, consider the categorical variable `Pclass` (passenger class). We can tabulate the categories and their corresponding observation counts as follows:

[5]:
```
df['Pclass'].value_counts().sort_index()
```

[5]:
```
Pclass
1    216
2    184
3    491
Name: count, dtype: int64
```

We now want to compute the averages of numerical variables (`Age`, etc.) separately for each passenger class. Instead of looping over categories, this can be achieved more elegantly with `groupby()`:

[6]:
```
# Group observations by accommodation class (first, second, third)
groups = df.groupby(['Pclass'])
```

The return value `groups` is a special pandas object which can subsequently be used to obtain group-specific data. To compute the group-wise averages, we can simply run

[7]:
```
groups.mean(numeric_only=True)
```

[7]:
```
        Survived        Age       Fare
Pclass
1       0.629630  38.233441  84.154687
2       0.472826  29.877630  20.662183
3       0.242363  25.140620  13.675550
```

Group objects support column indexing: if we want to only compute the total fare paid by passengers in each class, we can do this as follows:

[8]:
```
groups['Fare'].sum()
```

[8]:
```
Pclass
1    18177.4125
2     3801.8417
3     6714.6951
Name: Fare, dtype: float64
```

Built-in aggregations

There are numerous routines to aggregate grouped data, for example:

- `mean()`: compute average within each group
- `sum()`: sum values within each group
- `std()`, `var()`: within-group standard deviation and variance
- `median()`: compute median within each group
- `quantile()`: compute quantiles within each group
- `size()`: number of observations in each group
- `count()`: number of non-missing observations in each group
- `first()`, `last()`: first and last non-missing values in each column within each group
- `min()`, `max()`: minimum and maximum elements within a group

See the official documentation for a complete list.

Example: Number of elements within each group

We use `size()` and `count()` to compute group sizes or get the number of non-missing observations:

[9]:
```
groups.size()  # return number of elements in each group
```

[9]:
```
Pclass
1    216
2    184
3    491
dtype: int64
```

Note that `size()` and `count()` are two *different* functions. The former returns the group sizes as a `Series`, whereas `count()` returns the number of non-missing observations for *each* column:

[10]:
```
groups.count()  # return number of non-missing observations for each column
```

[10]:
```
        Survived  Name  Sex  Age  Ticket  Fare  Cabin  Embarked
Pclass
1            216   216  216  186     216   216    176       214
2            184   184  184  173     184   184     16       184
3            491   491  491  355     491   491     12       491
```

Example: Return first non-missing values in each group

We use `first()` to select the first non-missing value of each column within each group. These values can come from different rows. This is useful when a variable is constant within a group and we need to select a single value.

[11]:
```
# Return first non-missing value of each column within each group
groups[['Survived', 'Age', 'Sex', 'Fare']].first()
```

[11]:
```
        Survived   Age     Sex     Fare
Pclass
1              1  38.0  female  71.2833
2              1  14.0  female  30.0708
3              0  22.0    male   7.2500
```

Your turn

Use the Titanic dataset to perform the following aggregations:

1. Compute the average survival rate by sex (stored in the `Sex` column). You need to use the survival indicator stored in the column `Survived` for this.
2. Count the number of passengers aged 50+. Compute the average survival rate by sex for this group.
3. Count the number of passengers below the age of 20 by class and sex. Compute the average survival rate for this group by class and sex.

Writing custom aggregations

We can create custom aggregation routines by calling `agg()` on grouped data. This is an alias of `aggregate()` and applies custom functions to one column at a time.

For example, we can call the built-in aggregation functions listed above (`mean()`, `std()`, etc.) via `agg()`:

[12]:
```
# Calculate group means in needlessly complicated way
groups['Age'].agg('mean')

# More direct approach:
# groups['Age'].mean()
```

[12]:
```
Pclass
1    38.233441
2    29.877630
3    25.140620
Name: Age, dtype: float64
```

There is no advantage to doing this over directly calling `mean()` in this particular case, but `agg()` will come in handy if we need to apply *multiple* operations in a single call, as we'll see below.

We can also use `agg()` with a custom function. For example, we can compute the fraction of passengers aged 40+ in each class among those with known ages.

To understand what the final expression does, let's break it down into several steps. First, if we wanted to compute the fraction of passengers aged 40+ in the entire dataset, we could do this as follows:

[13]:
```
(df['Age'] >= 40).mean()
```

[13]:
```
np.float64(0.1829405162738496)
```

This suggests that the fraction is around 18%. However, this is not entirely correct, as it doesn't account for the fact that some passengers have missing ages. The next attempt drops the missing observations before computing the fraction:

[14]:
```
(df['Age'].dropna() >= 40).mean()
```

[14]:
```
np.float64(0.22829131652661064)
```

This variant reports the fraction among passengers with known ages, which is around 23%. We now want to compute this fraction for each passenger class. We can do this by grouping the data by `Pclass` and applying the same logic to each group. There is no built-in function to compute this fraction, so we have to define our own which accepts a `Series` of ages and returns the fraction. We then pass this function to `agg()`:

[15]:
```
# Define a custom aggregation function
def fcn(series):
```

```
    """
    Compute fraction of passengers aged 40+ among those with known ages.
    """
    return (series.dropna() >= 40).mean()


# Apply the custom aggregation function
groups['Age'].agg(fcn)
```

[15]:
```
Pclass
1    0.435484
2    0.213873
3    0.126761
Name: Age, dtype: float64
```

The shortest way to solve this task is to use a `lambda` expression to define the custom aggregation function in place. A `lambda` expression is basically a shorthand notation for defining a function in place without having to resort to a `def` statement.

[16]:
```
print('Fraction of passengers in each class aged 40+:')
# Apply a custom aggregation using a lambda expression
groups['Age'].agg(lambda series: (series.dropna() >= 40).mean())
```

```
Fraction of passengers in each class aged 40+:
```

[16]:
```
Pclass
1    0.435484
2    0.213873
3    0.126761
Name: Age, dtype: float64
```

Note that we called `agg()` only on the column `Age`; otherwise the function would be applied to every column separately, which is not what we want.

> **Your turn**
>
> Use the Titanic dataset to perform the following aggregation:
>
> - Group passengers by embarkation port (`Embarked`). Use `agg()` with a `lambda` expression to find the most common passenger class (`Pclass`) for each port. In statistics, the most common value is called the *mode* of a distribution.
>
> *Hint:* You can use the `mode()` method of a Series (e.g., `x.mode().iloc[0]`) or inspect the index of `value_counts()`.
> *Note:* When we group by `Embarked`, pandas excludes the two passengers whose embarkation port is missing.

Applying multiple functions at once

It is possible to apply multiple functions in a single call by passing a list of functions. These can be passed as strings or as callables (functions).

*Example: Applying multiple functions to a **single** column*

To compute the mean and median passenger age by class, we proceed as follows:

[17]:
```
groups['Age'].agg(['mean', 'median'])
```

[17]:
```
             mean  median
Pclass
1       38.233441    37.0
2       29.877630    29.0
3       25.140620    24.0
```

We can also pass functions directly. The NumPy functions `np.nanmean` and `np.nanmedian` ignore missing values:

[18]:
```
groups['Age'].agg([np.nanmean, np.nanmedian])
```

[18]:
```
          nanmean  nanmedian
Pclass
1       38.233441       37.0
2       29.877630       29.0
3       25.140620       24.0
```

The following more advanced syntax allows us to create new column names using existing columns and some operation:

```
groups.agg(
    new_column_name1=('column_name1', 'operation1'),
    new_column_name2=('column_name2', 'operation2'),
    ...
)
```

This is called "named aggregation" as the keywords determine the output column *names*.

*Example: Applying multiple functions to **multiple** columns*

The following code computes the average age and the highest fare in a single aggregation:

[19]:
```
groups.agg(average_age=('Age', 'mean'), max_fare=('Fare', 'max'))
```

[19]:
```
        average_age  max_fare
Pclass
1         38.233441  512.3292
2         29.877630   73.5000
3         25.140620   69.5500
```

Finally, `apply()` passes each group's `DataFrame` to a custom function, excluding grouping columns. It is useful for statistics that require values from several columns.

> **Your turn**
>
> Use the Titanic dataset to perform the following aggregations:
>
> 1. Compute the minimum, maximum, and average age by embarkation port (stored in the column `Embarked`) in a single `agg()` operation. There are several ways to solve this problem.
> 2. Compute the number of passengers, the average age, and the fraction of women by embarkation port in a single `agg()` operation. *Hint:* To compute the fraction of women, you can either use a `lambda` expression or first create a numerical indicator variable for females (as we did in the workshop).

1.2 Transformations

In the previous section, we combined grouping and reduction, i.e., data at the group level was reduced to a single statistic such as the mean. Alternatively, we can combine grouping with the `transform()`

function which assigns the result of a computation to each observation within a group and consequently leaves the number of observations unchanged.

For example, for *each* observation we could compute the average fare by class as follows:

[20]:
```
# Compute average fare by class, for each passenger
df.groupby('Pclass')[['Fare']].transform('mean').head(10)
```

[20]:
```
                Fare
PassengerId
1           13.675550
2           84.154687
3           13.675550
4           84.154687
5           13.675550
6           13.675550
7           84.154687
8           13.675550
9           13.675550
10          20.662183
```

As you can see, instead of collapsing the `DataFrame` to only 3 observations (one for each class), the number of observations remains the same, and the average fare is constant within each class.

We can then directly assign this average fare to a new column in the `DataFrame`:

[21]:
```
df['Avg_Fare'] = df.groupby('Pclass')[['Fare']].transform('mean')

# Print class, fare and average fare by class for first 10 passengers
df[['Pclass', 'Fare', 'Avg_Fare']].head(10)
```

[21]:
```
             Pclass     Fare   Avg_Fare
PassengerId
1                 3   7.2500  13.675550
2                 1  71.2833  84.154687
3                 3   7.9250  13.675550
4                 1  53.1000  84.154687
5                 3   8.0500  13.675550
6                 3   8.4583  13.675550
7                 1  51.8625  84.154687
8                 3  21.0750  13.675550
9                 3  11.1333  13.675550
10                2  30.0708  20.662183
```

As expected, the average fare is constant within each class, i.e., all passengers from class 1 are assigned the same average value, etc.

When would we want to use `transform()` instead of aggregation? Such use cases arise whenever we want to perform computations that include the individual value as well as an aggregate statistic.

Example: Deviation from average fare

Assume that we want to compute the relative difference between each passenger's fare and the average fare in their class, i.e.,

$$\text{Relative Difference}_i = \frac{\text{Passenger's Fare}_i}{\text{Average Fare in Class}_i} - 1$$

We could compute this using `transform()` as follows:

[22]:
```
# Compute relative difference of passenger's fare vs the avg. fare paid within class
avg_fare = df.groupby('Pclass')['Fare'].transform('mean')
df['Fare_Diff'] = df['Fare'] / avg_fare - 1

# Print relevant columns
```

```
df[['Pclass', 'Fare', 'Fare_Diff']].head(10)
```

[22]:

PassengerId	Pclass	Fare	Fare_Diff
1	3	7.2500	-0.469857
2	1	71.2833	-0.152949
3	3	7.9250	-0.420499
4	1	53.1000	-0.369019
5	3	8.0500	-0.411358
6	3	8.4583	-0.381502
7	1	51.8625	-0.383724
8	3	21.0750	0.541071
9	3	11.1333	-0.185897
10	2	30.0708	0.455354

Your turn

Use the Titanic dataset to perform the following transformation:

1. Compute the *excess* fare paid by each passenger compared to the minimum fare by embarkation port and class, i.e., compute $Fare - min(Fare)$ by port and class.

1.3 Resampling and aggregation

We discussed how to handle time series data in pandas in the previous lecture. This basically comes down to specifying an index which is a date or time stamp. Such an index allows us to easily perform operations such as computing leads, lags, and differences over time.

Another useful feature of the time series support in pandas is *resampling* which is used to group observations by time period and apply some aggregation function. This can be accomplished using the `resample()` method which in its simplest form takes a string argument that describes how observations should be grouped (`'YE'` for aggregation to years, `'QE'` for quarters, `'ME'` for months, `'W'` for weeks, etc.).

To illustrate, we load a dataset that contains daily observations on the value of the NASDAQ stock market index at close:

[23]:
```
# Path to NASDAQ data file
file = f'{DATA_PATH}/stockmarket/NASDAQ.csv'

# Read in NASDAQ data, set Date column as index
df = pd.read_csv(file, index_col='Date', parse_dates=True)

# Keep observations for 2025
df = df.loc['2025']

# Print first few rows
df.head()
```

[23]:

Date	NASDAQ
2025-01-02	19280.8
2025-01-03	19621.7
2025-01-06	19865.0
2025-01-07	19489.7
2025-01-08	19478.9

For example, we use `resample('ME')` to aggregate daily data to monthly frequency. This returns an object similar to the one returned by `groupby()`, on which we can call methods such as `mean()`:

[24]:
```
# Resample to monthly frequency, aggregate to mean of daily observations
# within each month
df.resample('ME').mean()
```

[24]:

Date	NASDAQ
2025-01-31	19565.045000
2025-02-28	19547.284211
2025-03-31	17828.028571
2025-04-30	16678.466667
2025-05-31	18642.361905
2025-06-30	19673.300000
2025-07-31	20784.900000
2025-08-31	21383.590476
2025-09-30	22188.600000
2025-10-31	23003.134783
2025-11-30	23023.626316
2025-12-31	23383.771429

Similarly, we can use `resample('W')` to resample to weekly frequency. Weeks end on Sunday and are labeled by that date. Below, we combine this with `last()` to return the last non-missing observation of each week:

[25]:
```
# Return last observation of each week, print first 10 rows
df.resample('W').last().head(10)
```

[25]:

Date	NASDAQ
2025-01-05	19621.7
2025-01-12	19161.6
2025-01-19	19630.2
2025-01-26	19954.3
2025-02-02	19627.4
2025-02-09	19523.4
2025-02-16	20026.8
2025-02-23	19524.0
2025-03-02	18847.3
2025-03-09	18196.2

Your turn

Use the daily NASDAQ data for 2025 and compute the percentage change from the first to the last trading day within each month.

1.4 List of functions used in this lecture

The following list contains the central functions covered in this lecture. Each function name is linked to the official API documentation which you can consult for more details regarding function arguments and return values. The [go to section] links to the relevant section in this notebook.

1.4.1 Grouping data

- `groupby()` — group data based on values of columns or index [go to section]

1.4.2 Built-in aggregation functions

Most of these methods can be applied to whole `DataFrame` or `Series` objects, or to objects obtained via `groupby()`.

- `mean()` — compute average within each group [go to section]
- `sum()` — sum values within each group [go to section]
- `std()`, `var()` — within-group standard deviation and variance [go to section]
- `median()` — compute median within each group [go to section]
- `mode()` — compute most frequent value(s) in a `Series` or `DataFrame` [go to section]
- `quantile()` — compute quantiles within each group [go to section]
- `size()` — number of observations in each group [go to section]
- `count()` — number of non-missing observations in each group [go to section]
- `first()`, `last()` — first and last non-missing values in each column within each group [go to section]
- `min()`, `max()` — minimum and maximum elements within a group [go to section]

1.4.3 Custom aggregation functions

- `agg()` — perform one or multiple (built-in or custom) aggregations *by column* [go to section]
- `apply()` — apply a custom function to each group's `DataFrame`, excluding grouping columns

1.4.4 Transformations

- `transform()` — perform (custom or built-in) transformation *by column* [go to section]

1.4.5 Aggregating time series data

- `resample()` — resample time series data to different frequency and apply aggregation [go to section]